the Copenhagen interpretation. What, after all, is a measurement? And how does it work? Wheeler never explains this, other than to insist that it has nothing to do with consciousness or life. Beyond that, he merely states that a measurement "is an irreversible act in which uncertainty collapses to certainty." Measurement, collapse—we're in familiar territory here, and Wheeler is in the usual trouble of needing to define what measurement is and how it happens, yet refusing to do exactly that. (Wheeler also states that the "essence" of quantum physics, "as the delayed-choice experiment shows, is *measurement*." But this is not particularly helpful for determining what actually constitutes a measurement.) There are, of course, other ways to look at this experiment—ways that are significantly different from Wheeler's ill-defined and questionably coherent ideas. Here are three of them.

Pilot-wave interpretation: A photon enters the beam splitter. Its pilot wave splits and travels down both paths, while the photon takes only one path (though we don't know which). If the second beam splitter isn't in place, the pilot wave reaches both detectors, carrying the particle along with it to one or the other.

If the second beam splitter is in place, the pilot wave interferes with itself when it arrives there, and never reaches detector 2. This prevents the photon from reaching detector 2 as well, no matter which path it traveled down.

It doesn't matter whether the second beam splitter is put in place before or after the photon goes through the first beam splitter—all that matters is whether the second beam splitter is there when the pilot wave arrives.

Many-worlds interpretation: A photon wave function enters the first beam splitter, splits, and travels down both paths. If the second beam splitter isn't in place, the photon wave function hits both of the detectors and entangles with the detectors' wave functions. Due to the enormous number of particles involved in this giant entangled wave function, decoherence occurs rapidly and the wave function branches. In one branch, the photon arrived at detector 1; in the other, it arrived at detector 2.

If the second beam splitter is in place, the photon wave function